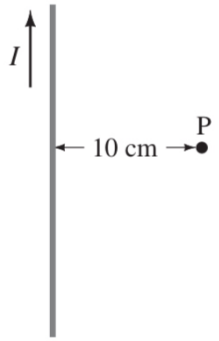
Magnetic field of long, straight wires

EXAMPLE 20–9 **Calculation of $\vec{B}$ near a wire.** An electric wire in the wall of a building carries a dc current of 25 A vertically upward. What is the magnetic field due to this current at a point P, 10 cm due north of the wire (Fig. 20–23)?

FIGURE 20–23 Example 20–9.



CAUTION

A compass, near a current, may not point north

APPROACH We assume the wire is much longer than the 10-cm distance to the point P so we can apply Eq. 20–6.

SOLUTION According to Eq. 20–6:

$$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A})(25\ \text{A})}{(2\pi)(0.10\ \text{m})} = 5.0 \times 10^{-5}\ \text{T},$$

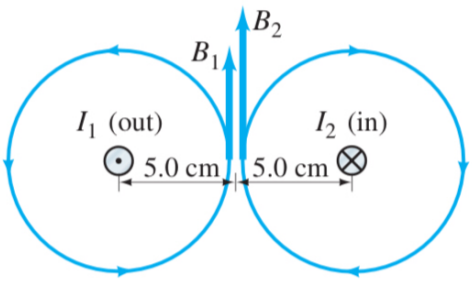
or 0.50 G. By right-hand-rule-1 (page 568), the field points to the west (into the page in Fig. 20–23) at point P.

NOTE The magnetic field at point P produced by the wire has about the same magnitude as Earth's, so a compass at P would not point north but to the northwest.

NOTE Most electrical wiring in buildings consists of cables with two wires in each cable. Since the two wires carry current in opposite directions, their magnetic fields cancel to a large extent, but may still affect sensitive electronic devices.

EXAMPLE 20–10 **Magnetic field midway between two currents.** Two parallel straight wires 10.0 cm apart carry currents in opposite directions (Fig. 20–24). Current $I_1 = 5.0$ A is out of the page, and $I_2 = 7.0$ A is into the page. Determine the magnitude and direction of the magnetic field halfway between the two wires.

FIGURE 20–24 Example 20–10. Wire 1 carrying current I_1 out towards us, and wire 2 carrying current I_2 into the page, produce magnetic fields whose lines are circles around their respective wires.



APPROACH The magnitude of the field produced by each wire is calculated from Eq. 20–6. The direction of *each* wire's field is determined with the right-hand rule. The total field is the vector sum of the two fields at the midway point.

SOLUTION The magnetic field lines due to current I_1 form circles around the wire of I_1, and right-hand-rule-1 (Fig. 20–8d) tells us they point counterclockwise around the wire. The field lines due to I_2 form circles around the wire of I_2 and point clockwise, Fig. 20–24. At the midpoint, both fields point upward in Fig. 20–24 as shown, and so add together. The midpoint is 0.050 m from each wire.

From Eq. 20–6 the magnitudes of B_1 and B_2 are

$$B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A})(5.0\ \text{A})}{2\pi(0.050\ \text{m})} = 2.0 \times 10^{-5}\ \text{T};$$

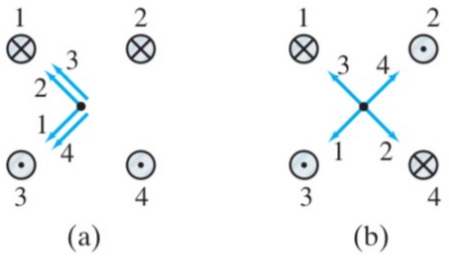
$$B_2 = \frac{\mu_0 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A})(7.0\ \text{A})}{2\pi(0.050\ \text{m})} = 2.8 \times 10^{-5}\ \text{T}.$$

The total field is *up* with a magnitude of

$$B = B_1 + B_2 = 4.8 \times 10^{-5}\ \text{T}.$$

CONCEPTUAL EXAMPLE 20–11 **Magnetic field due to four wires.** Figure 20–25 shows four long parallel wires which carry equal currents into or out of the page as shown. In which configuration, (*a*) or (*b*), is the magnetic field greater at the center of the square?

FIGURE 20–25 Example 20–11.



RESPONSE It is greater in (*a*). The arrows illustrate the directions of the field produced by each wire; check it out, using the right-hand rule to confirm these results. The net field at the center is the superposition of the four fields (which are of equal magnitude), which will point to the left in (*a*) and is zero in (*b*).

D1)[10]

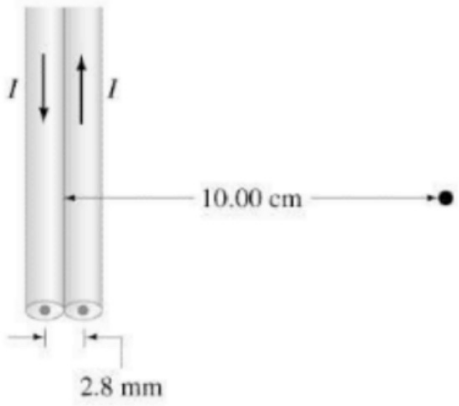
An overhead power line 8.5 meters above the ground carries 95A of current.

a) What is the magnitude of the magnetic field produced by the power line at ground level? How does it compare (percentage wise) to the Earth's magnetic field ($B_{Earth} \approx 50\mu T$)?
b) If the power line carries the current directly North-East (that is, $45°$ North of East), then which direction does the magnetic field vector at the ground point?

Answer: (a) $2.24\mu T$, (b) NW

D2)[11]

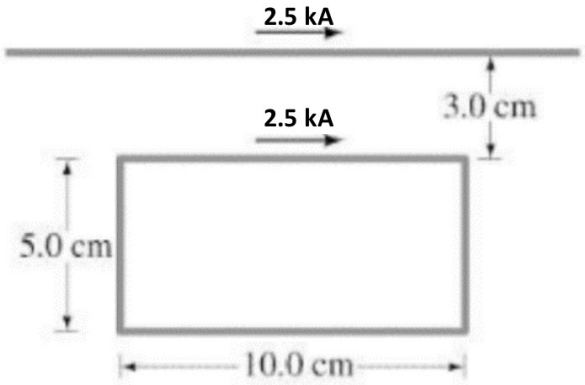
A long pair of wires conducts 25.0A of current to and from an instrument. The thin wires (dark grey) are separated by insulation (light grey) such that they are 2.8 mm apart. Consider a point 10 cm to the right of the midpoint between the two wires.



a) What is the total magnetic field at this point?
b) In which direction does the magnetic field point?

Answer: (a) $1.4\mu T$, (b) into page

D3)[12] A long, straight power line hangs above the ground and carries 2.5 kA (kiloamps) of current to the right. A rectangular loop of wire is oriented vertically and parallel to the power line, as shown. There is a current of 2.5 kA that flows clockwise in the loop.



a) What is the total magnetic force on the loop?

Hint: first find the magnetic field produced by the power line at the top and bottom of the loop, then find the force due to this field.

b) For what mass density of wire in grams per cm does the loop exactly "float" in the air?

Answer: (a) 2.61 N, (b) 8.87 g/cm

Lorentz force on a wire

EXAMPLE 20–1 **Magnetic force on a current-carrying wire.** A wire carrying a steady (dc) 30-A current has a length $\ell = 12\,\text{cm}$ between the pole faces of a magnet. The wire is at an angle $\theta = 60°$ to the field (Fig. 20–13). The magnetic field is approximately uniform at 0.90 T. We ignore the field beyond the pole pieces. Determine the magnitude and direction of the force on the wire.

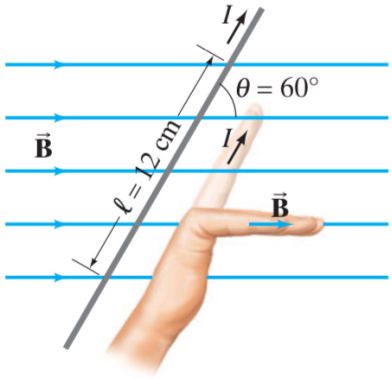


FIGURE 20–13 Example 20–1. For right-hand-rule-2, the thumb points into the page. See Fig. 20–11c.

APPROACH We use Eq. 20–1, $F = I\ell B \sin\theta$.

SOLUTION The force F on the 12-cm length of wire within the uniform field B is

$$F = I\ell B \sin\theta = (30\,\text{A})(0.12\,\text{m})(0.90\,\text{T})(\sin 60°) = 2.8\,\text{N}.$$

We use right-hand-rule-2 to find the direction of $\vec{\mathbf{F}}$. Hold your right hand flat, pointing your fingers in the direction of the current. Then bend your fingers (maybe needing to rotate your hand) so they point along $\vec{\mathbf{B}}$, Fig. 20–13. Your thumb then points into the page, which is thus the direction of the force F.

EXAMPLE 20–2 **Measuring a magnetic field.** A rectangular loop of wire hangs vertically as shown in Fig. 20–14. A magnetic field $\vec{\mathbf{B}}$ is directed horizontally, perpendicular to the plane of the loop, and points out of the page as represented by the symbol ⊙. The magnetic field $\vec{\mathbf{B}}$ is very nearly uniform along the horizontal portion of wire ab (length $\ell = 10.0\,\text{cm}$) which is near the center of the gap of a large magnet producing the field. The top portion of the wire loop is out of the field. The loop hangs from a balance (reads 0 when $B = 0$) which measures a downward magnetic force of $F = 3.48 \times 10^{-2}\,\text{N}$ when the wire carries a current $I = 0.245\,\text{A}$. What is the magnitude of the magnetic field B?

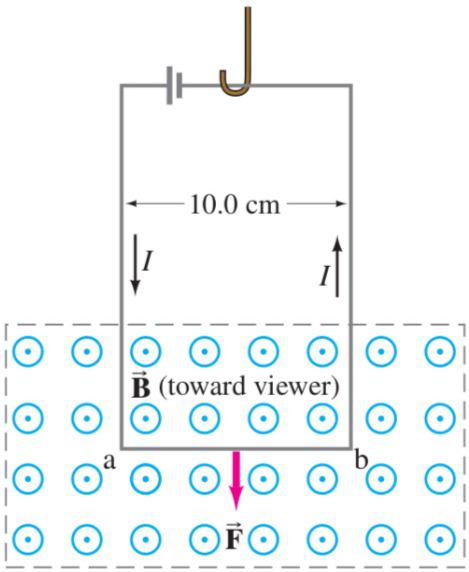


FIGURE 20–14 Measuring a magnetic field $\vec{\mathbf{B}}$. Example 20–2.

APPROACH Three straight sections of the wire loop are in the magnetic field: a horizontal section and two vertical sections. We apply Eq. 20–1 to each section and use the right-hand rule.

SOLUTION Using right-hand-rule-2 (page 564), we see that the magnetic force on the left vertical section of wire points to the left, and the force on the vertical section on the right points to the right. These two forces are equal and in opposite directions and so add up to zero. Hence, the net magnetic force on the loop is that on the horizontal section ab, whose length is $\ell = 0.100\,\text{m}$. The angle θ between $\vec{\mathbf{B}}$ and the wire is $\theta = 90°$, so $\sin\theta = 1$. Thus Eq. 20–1 gives

$$B = \frac{F}{I\ell} = \frac{3.48 \times 10^{-2}\,\text{N}}{(0.245\,\text{A})(0.100\,\text{m})} = 1.42\,\text{T}.$$

NOTE This technique can be a precise means of determining magnetic field strength.

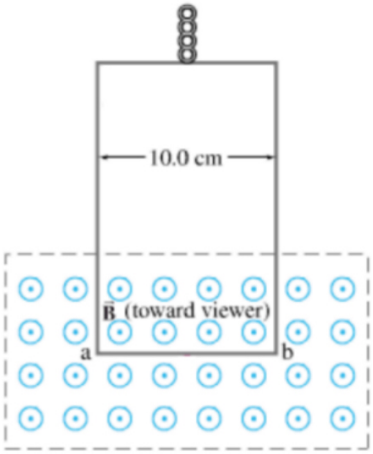
A1)[1]

A 160m long section of wire stretched between two towers is carrying a current of 150A. The earth's magnetic field of 5.0×10^{-5} T makes an angle of $65°$ with the wire. What is the magnitude of the magnetic force on the wire?

Answer: 1.09N

A2)[2]

A light loop of wire hangs from a chain such that its bottom in inside a region with a magnetic field of strength $B = 1.35T$. When a current runs in the loop, the loop pulls down on the chain with 5.5N of force.



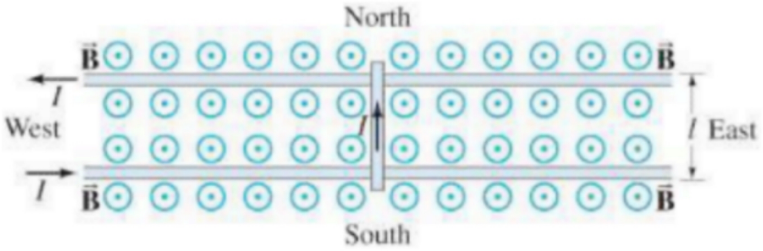
a) In which direction does the current run (clockwise or counterclockwise)?
b) How much current is flowing in the loop?

Note: the weight of the loop is negligible.

Answer: (b) 40.7A

A3)

A "rail gun" fires metal projectiles using magnetic forces. In the image, the horizontal rails are fixed, but the vertical piece of metal is free to slide and acts as the projectile.



The separation between the rails is 5cm and the magnetic field strength is 2.75T. A current of 6500A is run through the rails and projectile.

a) In which direction will the projectile fire (towards west or east in the figure)?
b) Compute the magnetic force on the projectile.
c) The projectile weighs 10g. Compute its acceleration.
d) The "barrel length" of the gun is 0.8 meters, such that the projectile is accelerated from rest over a 0.8-meter distance. At what speed will the projectile exit the rifle?

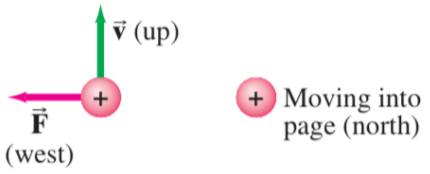
Hint: recall the kinematics equation $v_f^2 = v_i^2 + 2 \cdot a \cdot \Delta x$

Answer: (b) 893.75N, (d) 378m/s

Lorentz force on a charged particle

EXAMPLE 20–4 **Magnetic force on a proton.** A magnetic field exerts a force of 8.0×10^{-14} N toward the west on a proton moving vertically upward at a speed of 5.0×10^{6} m/s (Fig. 20–16a). When moving horizontally in a northerly direction, the force on the proton is zero (Fig. 20–16b). Determine the magnitude and direction of the magnetic field in this region. (The charge on a proton is $q = +e = 1.6 \times 10^{-19}$ C.)

FIGURE 20–16 Example 20–4.



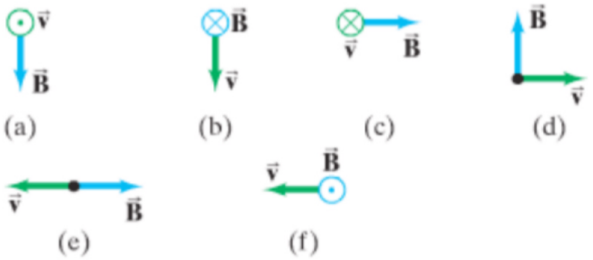
SOLUTION Equation 20–3 with $\theta = 90°$ gives

$$B = \frac{F}{qv} = \frac{8.0 \times 10^{-14}\ \text{N}}{(1.6 \times 10^{-19}\ \text{C})(5.0 \times 10^{6}\ \text{m/s})} = 0.10\ \text{T}.$$

APPROACH Since the force on the proton is zero when moving north, the field must be in a north–south direction ($\theta = 0°$ in Eq. 20–3). To produce a force to the west when the proton moves upward, right-hand-rule-3 tells us that $\vec{\mathbf{B}}$ must point toward the north. (Your thumb points west and the outstretched fingers of your right hand point upward only when your bent fingers point north.) The magnitude of $\vec{\mathbf{B}}$ is found using Eq. 20–3.

B1)[3] Right hand rule practice

A **negatively** charged particle moves in a magnetic field. In each image, the blue vector indicates the magnetic field and the green vector indicates the velocity). Find the direction of the force on the particle in each case.



Answers at top of next page

B2)[4] More right hand rule practice

A **positively** charged particle moves in a magnetic field. In each image, the green vector indicates the velocity, and the purple vector indicates the Lorenz force. Find the direction of the magnetic field in each case.



Answer (B1): (a) left, (b) left, (c) up, (d) into page, (e) no force, (f) down

Answer (B2): (a) down, (b) into-page, (c) right

B3)

During 1 second, 6.00×10^{20} electrons pass through a 3 cm length of wire. The wire is oriented perpendicular to a uniform magnetic field of strength 2.75 T. Compute the total force on the wire in two ways:

a) Compute the force on each individual electron using $F_M = q \cdot v \cdot B$ and add them up.
b) Compute the force on the current carrying wire using $F_M = \ell \cdot I \cdot B$

Answer: 7.93 N

B4)[5] A particle is accelerated from rest through a voltage of 15 kV and fired into a magnetic field of strength 4.00T directed perpendicular to the particle's velocity. Find the Lorenz force on each type of particle.

a) A proton ($q_p = 1.602 \times 10^{-19}$ C, $m_p = 1.673 \times 10^{-27}$ kg)
b) A deuteron (isotope of Hydrogen, important for nuclear fusion) ($q = q_p, m = 2m_p$)
c) An alpha particle (a.k.a. a Helium nucleus) ($q = 2q_p, m = 4m_p$)

Note:

- Recall that the electrostatic energy is $PE_{elec} = q \cdot U$ and kinetic energy is $KE = (1/2)\, m\, v^2$
- Use energy conservation: $PE_{elec} = KE$

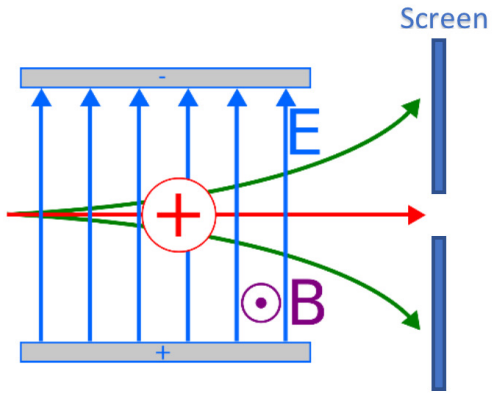
Answer: (a) 1.08 pN, (b) 0.77 pN, (c) 1.54 pN

where 1 piconewton (pN) = 10^{-12} N

B5)

Velocity Selector[6]

A "velocity selector" is a device that passes charged particles through uniform electric and magnetic fields which are perpendicular to each other to select particles with a particular velocity (see figure). A screen with a small open is placed to the right such that only particles that are **undeflected** will exit (red line). That means, the electric and magnetic forces on a particle must cancel out for it to pass the screen.



a. In which direction do the electric and magnetic forces act on a positively charged particle? On a negatively charged particle?
b. Find a formula for the velocity (v) of the exiting particles in terms of the electric and magnetic fields strength (E and B).
c. Recall that electric field has units of Newton/Coulomb and magnetic field has units of Tesla = Newton/(Ampere x Meter). Check that the units of your formula in part (b) make sense.
d. If the electric field has a fixed strength of 1.6×10^6 N/C, to which value must the magnetic field strength be adjusted to select particles traveling at 10% of the speed of light?

Note:

- Recall that the electrostatic force on a charge in an electric field is: $\vec{F}_e = q \cdot \vec{E}$

Answer: (d) 53.3 mT

Cyclotron

EXAMPLE 20–6 **Electron's path in a uniform magnetic field.** An electron travels at 1.5×10^7 m/s in a plane perpendicular to a uniform 0.010-T magnetic field. Describe its path quantitatively. Ignore gravity (= very small in comparison).

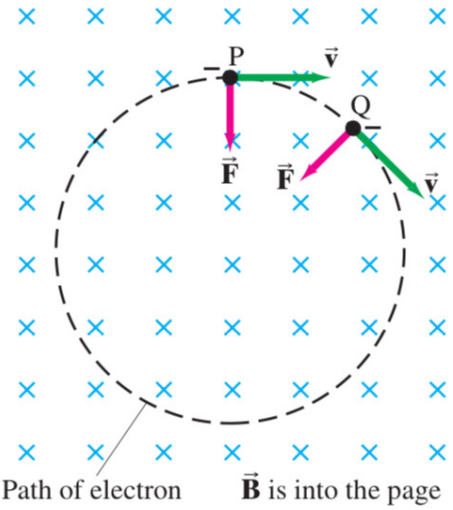


FIGURE 20–17 Force exerted by a uniform magnetic field on a moving charged particle (in this case, an electron) produces a circular path.

APPROACH The electron moves at speed v in a curved path and so must have a centripetal acceleration $a = v^2/r$ (Eq. 5–1). We find the radius of curvature using Newton's second law. The force is given by Eq. 20–3 with $\sin\theta = 1$: $F = qvB$.

SOLUTION We insert F and a into Newton's second law:

$$\Sigma F = ma$$
$$qvB = \frac{mv^2}{r}.$$

We solve for r and find

$$r = \frac{mv}{qB}.$$

Since $\vec{\mathbf{F}}$ is perpendicular to $\vec{\mathbf{v}}$, the magnitude of $\vec{\mathbf{v}}$ doesn't change. From this equation we see that if $\vec{\mathbf{B}}$ = constant, then r = constant, and the curve must be a circle as we claimed above. To get r we put in the numbers:

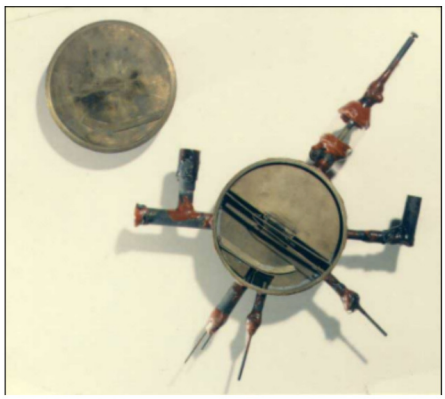
$$r = \frac{(9.1 \times 10^{-31}\,\text{kg})(1.5 \times 10^7\,\text{m/s})}{(1.6 \times 10^{-19}\,\text{C})(0.010\,\text{T})} = 0.85 \times 10^{-2}\,\text{m} = 8.5\,\text{mm}.$$

NOTE See Fig. 20–18. If the magnetic field B is larger, is the radius larger or smaller?

CONCEPTUAL EXAMPLE 20–7 **Stopping charged particles.** An electric charge q moving in an electric field $\vec{\mathbf{E}}$ can be decelerated to a stop if the force $\vec{\mathbf{F}} = q\vec{\mathbf{E}}$ (Eq. 16–5) acts in the direction opposite to the charge's velocity. Can a magnetic field be used to stop a charged particle?

RESPONSE No, because the force is always *perpendicular* to the velocity of the particle and thus can only change the direction but not the magnitude of its velocity. Also the magnetic force cannot do work on the particle (force and displacement are perpendicular, Eq. 6–1) and so cannot change the kinetic energy of the particle, Eq. 6–4.

C1)[7]



The first circular particle accelerator, the "cyclotron," created by Ernest Lawrence at Berkley in 1931, was 11 cm in diameter (yes, tiny! See image) and held **protons** ($m_p = 1.67 \times 10^{-27}$ kg, $q_p = 1.602 \times 10^{-19}$ C) in circular motion moving at a maximum speed of 3.93×10^6 m/s. What magnetic field was necessary to hold the protons in circular motion?

Answer: 0.745 T

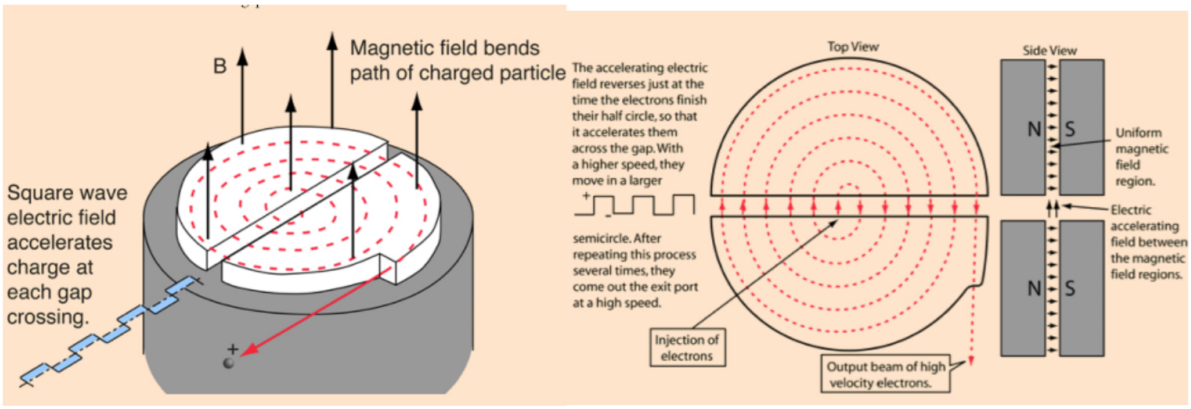
C2)



You would like to determine the magnetic field strength of some powerful neodymium magnets. To do this, you use a cathode ray tube (electron accelerator). The electrons ($m_e = 9.109 \times 10^{-31}$ kg, $|q_e| = 1.602 \times 10^{-19}$ C) are first accelerated through a voltage difference of 17'500 V. Subsequently, they pass near one of the neodymium magnets and are observed to move along a circular arc of radius 4 cm. What is the magnetic field strength in the region that the electrons are passing through?

Answer: 11.2 mT

C3)[8]



The idea behind a "cyclotron" is to accelerate particles in steps, by holding them in circular motion with a magnetic field and passing them through a voltage difference every half rotation. The polarity of the voltage must flip every half rotation so that the particle is always being accelerated (and not decelerated).

A certain cyclotron is accelerating protons ($m_p = 1.67 \times 10^{-27}$ kg, $q_p = 1.602 \times 10^{-19}$ C). The voltage difference between the two halves of the circle is 2700 V and the magnetic field strength is 0.15 Tesla.

a) What is the radius of proton's (semi)circular path after the first pass through the voltage difference?
b) What is the radius after the 2nd pass and the 3rd pass?
c) Recall for circular motion that the speed (v), period (T), and radius (R) are related by: $v = \frac{2\pi R}{T}$. Use this formula to compute the time (half period) it takes the particle to traverse the first, second, and third semicircle.

   You should find, amazingly, that the traversal time is always the same! That means that the voltage needs to be flipped in regular intervals to keep accelerating the particle.

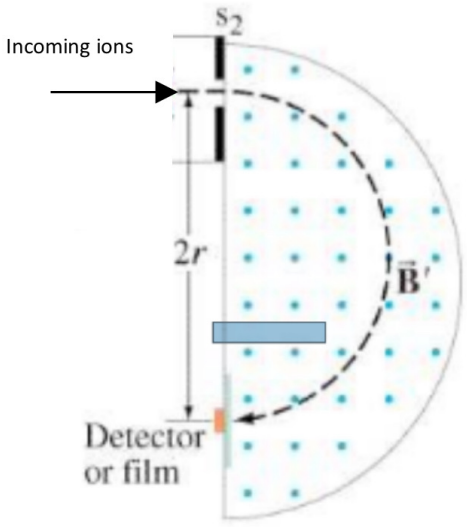
d) [**more challenging**] Derive a general formula for the rotational period of the cyclotron motion. It should only depend on the particle mass and charge and on the magnetic field.

Answer: (a) 5.00 cm, (b) 7.09 cm, 8.69 cm, (c) 0.22μs

C4)

Mass Spectrometer[9]

A mass spectrometer is a device used to determine the mass of atoms in a substance (and therefore which elements are contained therein). The substance is heated, causing the atoms to ionize; the ions are then fired into a region with a magnetic field, causing them to traverse a semicircle and be detected on a film (as shown to the right). Assume that the ions all have the same charge and have been passed through a velocity filter so that they enter the spectrometer with the same speed. (see B5 above).



A particular substance is known to contain carbon ($M_C = 12$u, where "u" is atomic mass units) and one other heavier element. If the film detects ions at a distance of $2r = 22.4$cm and $2r = 26.2$cm

What is the mass of the mystery element (in atomic mass units, u)?

What is the mystery element? (check the periodic table!)

Answer: 14u

Solenoid

E1)

A solenoid has a length, L, a number of loops, N, a cross sectional area A, and a current, I, that runs through the wire. Keeping everything else the same, what happens when:

a) The number of loops is tripled and the length is doubled.
b) The current is doubled and the cross sectional area is also doubled.
c) The current is quadruped (4x), the number of loops is halved, and the length is doubled.

Answer: (a) $B \rightarrow \frac{3}{2}B$, (b) $B \rightarrow 2B$, (c) $B \rightarrow B$

E2)

Compare the magnetic field strength in the interior of a solenoid with $n = 3$ loops/cm to that of a long, straight wire 2cm away from the wire when each carries 10A.

Answer: $0.1mT$, $3.8mT$

E3)

At a location on Earth where the Earth's magnetic field points exactly North, is parallel to the ground, and has a strength $B_{Earth} = 50\mu T$, a solenoid of length $L = 20cm$ is oriented with its long axis East/West. Current is run through the solenoid such that (in the interior) a magnetic field directed East is created.

a) If the solenoid has 100 loops 100mA of current is run through it, which direction will the total magnetic field point?
b) For a solenoid of the same length (20cm) and carrying the same current (100mA) how many windings should it have so that the total magnetic field points exactly North-East $(45°)$?

Answer: (a) $38.5°$ North of East, (b) 80 loops